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FA-Mamba: frequency attention driven Mamba for multimodal remote sensing classification

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ABSTRACT

Multimodal remote sensing classification plays a vital role in areas such as resource exploration and disaster monitoring. However, noise and redundancy across different modalities often hinder effective feature fusion. Existing approaches mainly rely on Transformer and Mamba architectures. While Transformer-based models suffer from high computational cost and inefficient long-range dependency modeling, Mamba improves efficiency but still focuses primarily on spatial-temporal domain modeling, with limited exploration of frequency-domain. This restricts the discriminative capacity of the fused representations. To overcome these limitations, we propose FA-Mamba, a frequency attention driven Mamba multimodal remote sensing classification network. With a focus on computational efficiency ensured by linear complexity, FA-Mamba adaptively enhances the complementary information between modalities while effectively suppressing noise and redundancy in cross-modal data, enabling more robust feature fusion. Additionally, to effectively capture global contextual information, we design a coordinate fusion attention (CFA) mechanism. By integrating spatial cues along different directions, the method incorporates long-range correlations across modalities. This facilitates a more comprehensive understanding of complementary multimodal information. Extensive experiments conducted on three public multimodal remote sensing datasets demonstrate the effectiveness of our approach. FA-Mamba achieves superior performance compared to state-of-the-art methods, with an average overall accuracy of 95.84%. Our code is available at <https://github.com/dnyeee/FA-Mamba>.

1. Introduction

The deep integration of remote sensing technology and deep learning has ushered land cover classification into a new era of intelligence (Wang et al., 2022; Xie et al., 2020; Xiong et al., 2022). With the continuous advancement of Convolutional Neural Networks (CNNs), single-modality remote sensing data can now achieve remarkably high accuracy rates in classification tasks (Wu et al., 2024). However, in practical applications involving complex and diverse geographic environments, the inherent physical limitations of single-modality data have become a major obstacle to further technological breakthroughs. For instance, Hyperspectral imaging (HSI) provides fine-grained material identification thanks to its high spectral resolution (Fan et al., 2017), while Light Detection and Ranging (LiDAR) is highly effective at capturing precise 3D structural information of both terrain and surface objects (Wang & Menenti, 2021). Nonetheless, HSI is susceptible to atmospheric interference and performs poorly under adverse weather conditions,

whereas LiDAR lacks spectral richness, often resulting in increased material misclassification rates. These modality-specific limitations significantly hinder the stability and classification accuracy of single-source systems, particularly when operating under variable weather conditions and complex environmental contexts, thereby making the fusion of multi-modal data indispensable for building a unified sensing framework to surpass the information bottlenecks imposed by single-modality data (Hong et al., 2020). By integrating complementary sensing modalities, multimodal approaches can substantially enhance robustness, classification accuracy, and generalization capabilities in a wide range of Earth observation tasks. Such fusion enables a more comprehensive and holistic understanding of scenes, compensating for information gaps introduced by individual sensor limitations or challenging environmental conditions. With ongoing advancements in spaceborne sensor technology, multimodal classification has achieved remarkable progress in multiple application domains, including precision agriculture for crop health assessment, geological interpretation in mining regions, urban

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3D modeling, forest biomass estimation, and flood disaster assessment. Despite these advancements, multimodal land-cover classification still encounters unresolved challenges stemming from cross-modal noise and modality inconsistency, which can negatively affect the effectiveness of feature alignment and fusion. Consequently, the development of robust fusion strategies capable of adaptively filtering out noise while fully leveraging the complementary strengths of multiple modalities remains a critical and active area of research. Therefore, multimodal fusion must be able to withstand cross-sensor noise and heterogeneity while also scaling to high-resolution images. This is one of the direct motivations behind our architecture. From a modeling perspective, existing studies mainly follow the CNN, Transformer, and SSM-based paradigms, reviewing their respective strengths and limitations helps motivate our design choices.

In recent years, CNNs dominate multimodal classification by extracting local spatial-spectral features efficiently. However, their limited receptive fields struggle to model long-range dependencies, and pooling operations degrade high-frequency details. While Transformers address these issues via global self-attention and multi-head mechanisms for cross-modal feature fusion (Ma et al., 2024b), their quadratic computational complexity hinders deployment on high-resolution remote sensing imagery. Against this backdrop, state space model (SSM), originally derived from the classical Kalman filter (Kalman, 1960), have emerged as a novel technical pathway for multimodal remote sensing classification, particularly with the introduction of the Mamba architecture (Gu & Dao, 2023). Mamba employs a selective state space mechanism that compresses previously scanned information into hidden states, enabling dynamic receptive field adjustment while maintaining linear computational complexity. Its linear complexity is due to the state update of selective scan: the model compresses historical information with a fixed-dimensional hidden state during the token scanning process, and performs causal recursive update only once for each token. Therefore, the overall computational cost grows approximately linearly with the number of tokens, without the need for secondary pairwise interaction computation as required by self-attention. This data-dependent state transition mechanism inherits the global modeling advantages of Transformers while mitigating their excessive computational resource consumption. Recently, Mamba has gained prominence across various domains due to its efficient sequence modeling capabilities, demonstrating strong potential as a replacement for Transformer architectures. However, when applied to Earth observation tasks involving wide-swath and high-resolution remote sensing imagery, the standard Mamba model exhibits clear limitations. Specifically, its state space-based modeling mechanism shows insufficient sensitivity to fragmented object boundaries commonly found in high-resolution imagery. Moreover, the use of fixed-scale convolution kernels makes it difficult to adaptively represent multi-scale surface features (Elizar et al., 2022), limiting the model's expressiveness in dynamically structured environments. These structural limitations have hindered Mamba from achieving technological breakthroughs in many related tasks. These structural limitations have hindered Mamba from achieving technological breakthroughs in many related tasks. Furthermore, existing approaches generally prioritize spatial and temporal domain modeling, while the latent discriminative and complementary characteristics embedded in the frequency domain across different modalities remain largely underexplored, ultimately constraining the richness and effectiveness of multimodal feature representations.

In contrast, traditional spatial or temporal domain methods mainly optimize local state transitions and face two key limitations: (1) their inherently limited receptive fields often fail to capture critical global structures such as texture and edge patterns, which are essential for fine-grained discrimination in remote sensing imagery, and they also tend to overlook cross-modal noise and interference, further compromising the effectiveness of multimodal feature fusion; (2) spatial-domain modeling of long-range dependencies typically relies on computationally intensive operations like large-kernel convolutions or deep stack-

ing, which restrict scalability on high-resolution data. To address these limitations, we introduce an adaptive frequency attention mechanism that incorporates a spectral transformation pathway after state space modeling. This transformation projects features into the frequency domain, enabling the model to capture global dependencies through dynamic receptive fields without adding extra parameters. By aggregating multi-scale frequency patterns in a compact and learnable manner, and applying a data-driven spectral weighting strategy to emphasize informative frequency bands while suppressing irrelevant or noisy signals, this design enhances feature discriminability. Additionally, the use of efficient frequency-domain operations substantially reduces both memory consumption and computational costs in long-range modeling, outperforming conventional spatial-domain approaches. Building upon these insights, we propose FA-Mamba, a frequency attention driven Mamba multimodal remote sensing classification network. FA-Mamba captures long-range dependencies in multimodal data through a lightweight yet effective architecture centered on the Mamba Driven Multimodal Fusion (M2F) encoder. The M2F encoder consists of two core components: the frequency attention driven Mamba (FAM) module, which adaptively strengthens frequency-domain representations, models global dependencies, and mitigates cross-modal interference to improve the effect of fused features; and the coordinate fusion attention (CFA) mechanism, which extracts global structural information to further improve cross-modal feature fusion. Our contributions can be summarized as follows:

- We propose FA-Mamba, a frequency attention driven Mamba multimodal remote sensing classification network. Under the constraint of linear complexity, it adaptively enhances complementary information across modalities while mitigating cross-modal interference.
- We propose a global feature extraction based on a coordinate fusion attention mechanism, CFA mechanism. It captures spatial positional information using pooling operations along two directions and integrates long-range dependencies to fully exploit complementary multimodal information, thereby improving classification accuracy.
- Comprehensive experiments conducted on three widely recognized benchmark datasets for HSI and LiDAR classification clearly demonstrate that our method consistently achieves state-of-the-art performance, across diverse and challenging remote sensing scenes.

2. Related work

2.1. Multimodal classification

Traditional multimodal fusion approaches commonly include sub-pixel-level fusion and pixel-level fusion. For instance, Gao et al. achieves data integration based on differences in spatial resolution (Gao et al., 2006), while Amorós-López et al. directly fuses aligned pixel data (Amorós-López et al., 2013), a technique frequently used for multi-sensor data from the same platform. With the advancement of deep learning, an increasing number of technologies have emerged, giving rise to more sophisticated multimodal fusion networks to tackle different challenges. To address adversarial learning problems, Lu et al. proposed the CALC framework (Lu et al., 2023), which employs a coupled strategy that enhances classification performance while effectively managing computational complexity. To mitigate feature redundancy, Xie et al. proposed the ECNet (Xie et al., 2024), which combines CNNs and Transformers, leveraging the Ebbinghaus memory curve to optimize multi-source data fusion. Additionally, leveraging foundation models, Ma et al. introduced MANet (Ma et al., 2024a), which integrates a Multimodal Adapter (MMAdapter) into the Segment Anything Model (SAM) to improve generalization across optical and elevation data. Although previous methods have made progress in multimodal remote sensing classification, they still have limitations in effectively modeling long-range dependencies and leveraging frequency domain characteristics. In contrast, FA-Mamba employs a SSM to efficiently model long-range dependencies with linear computational complexity, making it well-suited for

large-scale remote sensing applications. Additionally, by incorporating Fourier frequency feature extraction, FA-Mamba adaptively emphasizes key frequency components, thereby enhancing feature discriminability. This synergistic combination enables FA-Mamba to achieve a balance between efficiency and accuracy.

2.2. State space model

State Space Models, originating from Kalman filters (Kalman, 1960), have emerged as a powerful framework for modeling long-range dependencies in sequential data with linear computational complexity. The modern SSM architecture gained prominence through the S4 model (Gu et al., 2021), which introduced structured state matrices and HiPPO initialization to effectively capture long-term temporal dynamics. Subsequent innovations like the S5 model (Smith et al., 2022) expanded SSM capabilities by enabling multi-input multi-output (MIMO) mechanisms and parallel scan strategies. The advent of Mamba (Gu & Dao, 2023) marked a breakthrough through its selective scanning mechanism (S6), which dynamically adjusts parameters based on input content while maintaining hardware efficiency and global context modeling. This innovation enabled SSMs to surpass Transformer-based models in language tasks (Gu & Dao, 2023) and inspired broader computer vision applications.

In computer vision, early SSM variants addressed critical challenges: Zhu et al. proposed Vim (Zhu et al., 2024) to improve positional awareness in visual sequence modeling, using bidirectional SSMs and positional embeddings. Meanwhile, Liu et al. proposed VMamba (Liu et al., 2024b), which develops the 2D Selective Scanning (SS2D) module to extend 1D sequence modeling to non-sequential image data through four-directional scanning paths. Both approaches achieve efficient visual feature learning with linear complexity. Later advancements like LocalMamba (Huang et al., 2024) proposed by Huang et al. to further optimize spatial-adaptive scanning, and RSMamba (Chen et al., 2024) introduced by Chen et al. to integrate local-global feature interactions. For hyperspectral image classification, Liu et al. developed HyperMamba (Liu et al., 2024a) to combine spectral-spatial adaptive scanning with frequency domain enhancement for addressing remote sensing challenges. In recent years, SSMs have also been explored in low-level vision tasks: Li et al. proposed FourierMamba (Li et al., 2024) to leverage frequency-domain transformations for optimizing image reconstruction, while Zhuang et al. designed FAHM (Zhuang et al., 2025) to unify spatial-spectral modeling through hierarchical state transitions. Additionally, in sequential recommendation systems, Wang et al. created EchoMamba (Wang et al., 2024) to utilize bidirectional state propagation and frequency-aware filtering for modeling user behavior dynamics. The flexibility of SSMs is further demonstrated in biomedical imaging, where Zheng et al. established FDVM-Net (Zheng & Zhang, 2024) to combine the local feature extraction capabilities of convolutional layers with SSMs' long-range dependency capturing ability, thereby achieving high-quality, clear images. In contrast, FA-Mamba couples SS2D-based long-range modeling with frequency attention and coordinate-aware fusion, which jointly enhance modality complementarity while preserving linear-complexity inference. However, most existing SSM variants primarily focus on single-modality representation learning, and do not explicitly address cross-modal interference. In contrast, FA-Mamba couples SS2D-based long-range modeling with frequency attention and coordinate-aware fusion, which jointly enhance modality complementarity while preserving linear complexity inference.

2.3. Fast Fourier transform

Fast Fourier Transform-based methods have become indispensable tools for global context modeling and cross-domain feature learning in deep learning. Renowned for decomposing spatial signals into frequency components, FFT provides a computationally efficient framework for

simultaneous analysis of global patterns and local details (Xiao et al., 2024). Early applications in vision tasks, including semantic segmentation (Yang & Soatto, 2020) and image classification (Xu et al., 2021), demonstrated its potential in capturing holistic statistical signatures. In low-level vision, Mao et al. (2021) developed residual FFT modules to enhance high-frequency details in low-light enhancement, while Li et al. (2023) employed FFT to mitigate noise amplification during luminance adjustment.

Subsequent research has expanded FFT applications through innovations like window-based frequency channel attention mechanisms (Guo et al., 2023) and joint frequency-spatial learning for super-resolution tasks (Wang et al., 2023). Crucially, distinct information manifests as specific frequency components in the spectral domain, enabling FFT-based processing to achieve superior discriminative capability compared to physical space operations (Zhang et al., 2025). In hyperspectral image classification, Shi et al. (2023) pioneered F3Net, combining FFT with convolutional tokenizers and learnable spectral filters to model spectral-spatial interactions. Advanced architectures like FAHM (Zhuang et al., 2025), proposed by Wang et al., integrate FFT with state space models for hierarchical feature optimization. Recently, Chang et al. introduced EM-Net (Chang et al., 2024) to automatically amplify task-relevant frequency components while suppressing noise, thereby achieving state-of-the-art performance across multiple benchmarks.

3. Method

This section introduces the overall architecture of our proposed FA-Mamba framework, which is designed for multimodal remote sensing classification by effectively integrating HSI and LiDAR data. The framework adopts a state space driven design to enhance local dependency modeling, while leveraging frequency-domain attention to capture complementary modality-specific features. In the following subsections, we present detailed descriptions of components, including the FAM Module and the CFA mechanism, which jointly contribute to effective multimodal feature representation. This design is motivated by two practical needs in HSI-LiDAR fusion: capturing long-range spectral dependencies with low computational overhead, and suppressing cross-modal redundancy under modality mismatch and noise. Accordingly, we pair an SSM backbone for efficient global context modeling with frequency-aware and coordinate-aware interactions to balance complementary cues across modalities.

3.1. Overall architecture

As depicted in Fig. 1(a), our proposed FA-Mamba facilitates multimodal remote sensing image classification through three pivotal stages: feature extraction and preprocessing, a Mamba Driven Multimodal Fusion Encoder, and final prediction and classification. The model employs a dual-branch network architecture to process HSI and LiDAR data. During the feature extraction and preprocessing stage, both HSI and LiDAR inputs undergo modality-specific processing through standardized CNN blocks. Each block integrates a convolutional feature extractor with spectral-spatial normalization layers and nonlinear activation, followed by progressive downsampling operations to generate discriminative embeddings (denoted as F_H and F_L for HSI and LiDAR respectively). This modular design ensures stable gradient propagation while optimizing training convergence and generalization. Subsequent hierarchical processing introduces stochastic regularization through Dropout layers to counteract overfitting. The dual-branch network maintains independent parameter spaces until later fusion stages, thereby preserving modality-specific characteristics during feature refinement.

Following preprocessing, HSI and LiDAR feature embeddings (F_H and F_L) undergo statistical distribution alignment to ensure compatibility for downstream processing. These normalized representations are then fed into FAM module, which synergizes Mamba's linear-complexity long-range dependency modeling with spectral domain analysis. Within

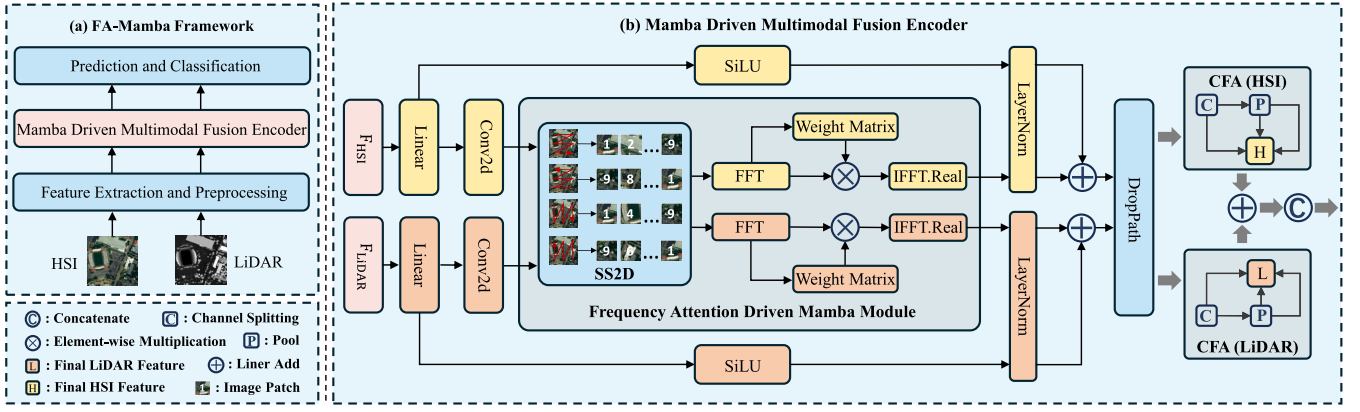


Fig. 1. Overall framework of FA-Mamba, which includes: (a) Overall FA-Mamba Framework, which shows the overall flow of the whole network, including three steps, feature extraction and preprocessing, Mamba Driven Multimodal Fusion encoder, and prediction and classification. (b) Implementation details of Mamba Driven Multimodal Fusion encoder, which contains two important modules, Frequency Attention Driven Mamba module and Coordinate Fusion Attention mechanism.

FAM, the module dynamically recalibrates feature importance by adaptively amplifying task-relevant spectral components and suppressing noise. A random path sampling regularization mechanism is then applied to enhance the model's robustness. The refined outputs (Y_H and Y_L) undergo a series of state-space-guided feature evolution, frequency-domain processing, and regularization, ultimately preserving global contextual information while adaptively highlighting key frequency-domain features. This process is mathematically formalized through an additive feature fusion formulation, which preserves the inherent global contextual integrity of the state space paradigm. The specific formulation is as follows:

$$Y_H = F_H + \text{FAM}(F_H), \quad (1)$$

$$Y_L = F_L + \text{FAM}(F_L). \quad (2)$$

The refined feature embeddings (Y_H and Y_L) from the FAM-based residual structure are then fed into the CFA mechanism, which strategically integrates multi-scale contextual information through a dual-branch architecture. Furthermore, we intentionally place the CFA after the FAM to address the imbalance between local and global representations. FAM first refines the modality-specific features by extracting fine-grained local spectral-spatial cues and suppressing redundant responses via frequency recalibration. CFA then focuses on capturing global spatial dependencies and coordinate-aware interactions, enabling more stable cross-modal fusion than applying global spatial modeling on raw features. CFA mechanism mitigates these limitations through multi-scale extraction and adaptive cross-modal interaction. Its detailed structure will be described in later sections. Then, a residual-enhanced feature concatenation is performed to generate a unified feature representation, which explicitly preserves cross-modal complementary information while maintaining the fidelity of original features through a cascaded structure. The mathematical formulation of this process is as follows:

$$F_{\text{fused}} = C[(Y'_H + Y'_L), Y'_H, Y'_L], \quad (3)$$

where C is the feature fusion operation, Y'_L and Y'_H denotes the feature after the processing of the CFA module. Finally, this fused representation is passed to a classification layer, producing the ultimate classification output.

In our model, we utilize the cross-entropy loss function to measure the difference between the predicted class probabilities and the true class labels. This loss function is especially effective for multi-class classification tasks due to its ability to impose a stronger penalty when the model assigns a low probability to the correct class. For a given sample,

the cross-entropy loss is computed using the following formula:

$$\text{Loss} = - \sum_{c=1}^M y_c \log(p_c).$$

Here, M represents the total number of classes, y_c is a binary indicator (taking the value 1 if class c is the correct label for the sample, and 0 otherwise), and p_c denotes the model's predicted probability that the sample belongs to class c . The training process optimizes the model parameters by minimizing this cross-entropy loss, employing the Adam optimizer to iteratively refine the parameters and reduce the loss value. Throughout training, we track the loss on both the training and validation sets to confirm effective learning and to prevent overfitting during model selection. This choice of loss function aligns closely with our task's objectives. By evaluating the divergence between the predicted and actual probability distributions, the cross-entropy loss offers a precise optimization target, thereby improving the model's classification accuracy.

3.2. Frequency attention driven Mamba module

As illustrated in Fig. 1(b), the M2F encoder contains two core components, the FAM module and the CFA mechanism. Among them, the FAM module is primarily designed to efficiently handle ultra-long spectral sequences and perform adaptive feature enhancement from a frequency domain perspective. This module consists of two key components: 1) SS2D module, which extracts temporal features and joint spectral-spatial features from HSI and LiDAR data; 2) frequency attention mechanism, which transforms features into the frequency domain for processing. Additionally, the entire module adopts a residual structure to prevent gradient vanishing while further enhancing the semantic segmentation performance of the SSM.

Specifically, the architecture of the FAM module begins with a linear transformation layer. We take the preprocessed HSI feature F_H as an illustrative example, and note that the processing of LiDAR features follows a similar procedure. After a brief transformation through the linear layer, F_H is passed into the SS2D module, which is responsible for efficient feature extraction and modeling of long-range dependencies. The SS2D module is specifically designed to accommodate multimodal inputs (such as HSI and LiDAR), and effectively captures dependencies along both spatial and spectral dimensions. Its internal architecture consists of three key components: scan expanding, the S6 block, and scan merging. The scan expanding operation unfolds the input feature map along multiple directional axes, thereby converting the original 2D image into several 1D sequences. Let $\{F_H^{(1)}, F_H^{(2)}, \dots, F_H^{(d)}\}$ denote the collection of 1D sequences obtained by expanding the original 2D feature representation F_H along d distinct paths. For each path $k \in 1, 2, \dots, d$,

a corresponding sequence is constructed by traversing pixels along a predefined directional path:

$$F_H^{(k)} = \text{expand}_k(F_H), \quad (4)$$

and

$$F_H^{(k)}[t] = F_H[i, j], \quad (5)$$

where $[i, j]$ represents the pixel position in the original 2D feature map that corresponds to the t -th position in the expanded sequence $F_H^{(k)}$. Subsequently, the S6 block is applied to the unfolded sequences to extract features, capturing global information in various directions. The core of S6 is the SSM, used to model long-term dependencies, with its state update formula as follows:

$$h'(t) = Ah(t) + Bx(t), \quad (6)$$

$$y(t) = Ch(t) + Dx(t), \quad (7)$$

where matrix A determines the temporal evolution of hidden states, matrix B maps input signals to state updates, matrix C transforms hidden states into observable outputs and matrix D enables direct feedthrough from inputs to outputs. Since images are discrete signals, Mamba discretizes the continuous equation using the zero-order hold method, with a discretization step size Δ , yielding the discrete parameters $\bar{A}$ and $\bar{B}$:

$$\bar{A} = e^{\Delta A}, \quad (8)$$

$$\bar{B} = (\Delta A)^{-1}(e^{\Delta A} - I) \cdot \Delta B. \quad (9)$$

Thus, the discretized state update formula is:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad (10)$$

$$y_t = Ch_t + Dx_t. \quad (11)$$

In S6, Δ , B, and C are input-dependent rather than fixed, allowing the model to adaptively adjust the state transition based on the current input, thereby more flexibly capturing key information in the sequence. Integrate the above series of SSM operations into the S6 transformation and apply it to the every 1D expanded sequence in each direction:

$$F_H'^{(k)} = S6(F_H^{(k)}), \quad (12)$$

where $F_H'^{(k)}$ denotes the feature representation processed by the S6 block. Subsequently, the outputs of S6 obtained from the d different directional paths are aggregated together in order to capture globally coherent features:

$$F_H' = \sum_d \text{Unflatten}_k(F_H'^{(k)}), \quad (13)$$

where $\text{Unflatten}_k(\cdot)$ is the operation that maps the expanded sequence back to the two-dimensional structure following the original path. The final fused feature has the same dimensions as the original input feature F_H , i.e., $F_H' \in \mathbb{R}^{B \times C \times H \times W}$.

Extracting frequency characteristics of features often reveals deeper properties of the image. The intensity and distribution of different frequency components in an image reflect its texture, edges, and shapes, which are crucial for classification tasks. The 2D Discrete Fourier Transform (DFT), as a cornerstone of digital signal processing, is a commonly used method for converting features into the frequency domain. Its formula can be expressed as:

$$X[k, l] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} x[m, n] \cdot e^{-j2\pi \left(\frac{km}{M} + \frac{ln}{N} \right)}, \quad (14)$$

$X[k, l]$ is frequency domain signal, indexed by k (horizontal, $0 \leq k < M$) and l (vertical, $0 \leq l < N$); $x[m, n]$ is spatial domain signal, indexed by m (horizontal, $0 \leq m < M$) and n (vertical, $0 \leq n < N$). Fourier Transform has a very important property, known as the Convolution Theorem. The Convolution Theorem states that the Fourier Transform of the convolution of two signals is equal to the product of their Fourier Transforms. This implies that the convolution operation can be transformed

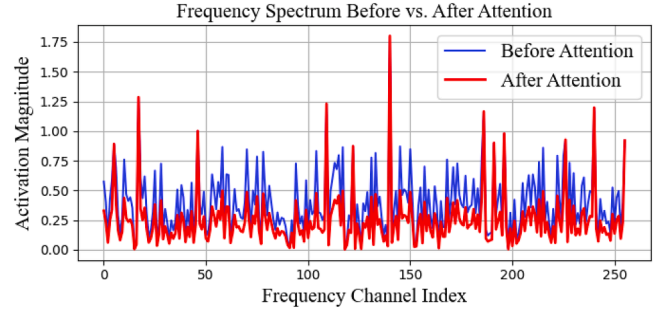


Fig. 2. Effect of frequency attention on feature representation. The figure shows the spectrum comparison of the features before and after applying the frequency attention mechanism.

into pointwise multiplication in the frequency domain, significantly reducing computational complexity. The above theorem can be expressed as the following formula:

$$\text{DFT}\{x \otimes y\}[k, l] = \text{DFT}\{x\}[k, l] \cdot \text{DFT}\{y\}[k, l], \quad (15)$$

where $\otimes$ is circular convolution operation (2D).

In addition, we use FFT to compute the Discrete Fourier Transform (DFT), which reduces the computational complexity from $O(N^2)$ to $O(N \log N)$. Benefiting from these properties, FFT offers significant computational advantages when processing large-scale remote sensing images. The frequency attention mechanism in the FAM module fully leverages these characteristics to achieve adaptive global receptive field adjustment without requiring additional parameters. Through the effect of a learnable weight matrix W , task-relevant frequency components are enhanced, while redundant low-frequency noise is suppressed. Specifically, W is formed by performing regional pooling on the frequency domain features and then recombining the pooled features, thereby generating a compact global spectral matrix that can stably reweight the spectrum and protect it from local noise. Fig. 2 shows the effect of frequency attention on feature representation. The figure shows that after frequency attention, the curve exhibits a significantly reduced activation amplitude in most frequency channels. It shows that frequency attention suppresses redundancy and noise, which makes the feature compact enhanced. In contrast, the key frequency components are enhanced, which highlights the ability of the attention module to enhance important information.

As shown in Fig. 1(b), the features output by the SS2D module are transformed to the frequency domain via FFT, multiplied element-wise by a weight matrix W , and then converted back to the time domain via inverse fast fourier transform (IFFT) followed by taking the real part. The operation of extracting the real part effectively removes the imaginary noise artifacts introduced during frequency-domain processing, while simultaneously maintaining the physical interpretability and semantic clarity of the enhanced feature representations. This process can be expressed as:

$$F_H'' = \text{Re} \left(F^{-1} \left(F(F_H') \odot W \right) \right), \quad (16)$$

where F denotes FFT, transforming the signal to frequency domain, $\odot$ represents element-wise multiplication in the frequency domain, F^{-1} denotes the IFFT, transforming the signal back to the time domain, and $\text{Re}(\cdot)$ denotes taking the real part of the complex result. The weight matrix W is derived based on the frequency characteristics of the original feature itself, highlighting key frequency features to enhance the expressive power of the features. First, the frequency characteristics are divided into real and imaginary parts, then global average pooling is applied to each, and finally, the two parts are combined to form the weight matrix W , expressed as:

$$W = \text{GAP} \left[\text{Re}(F_H'') \right] + j \text{GAP} \left[\text{Im}(F_H'') \right]. \quad (17)$$

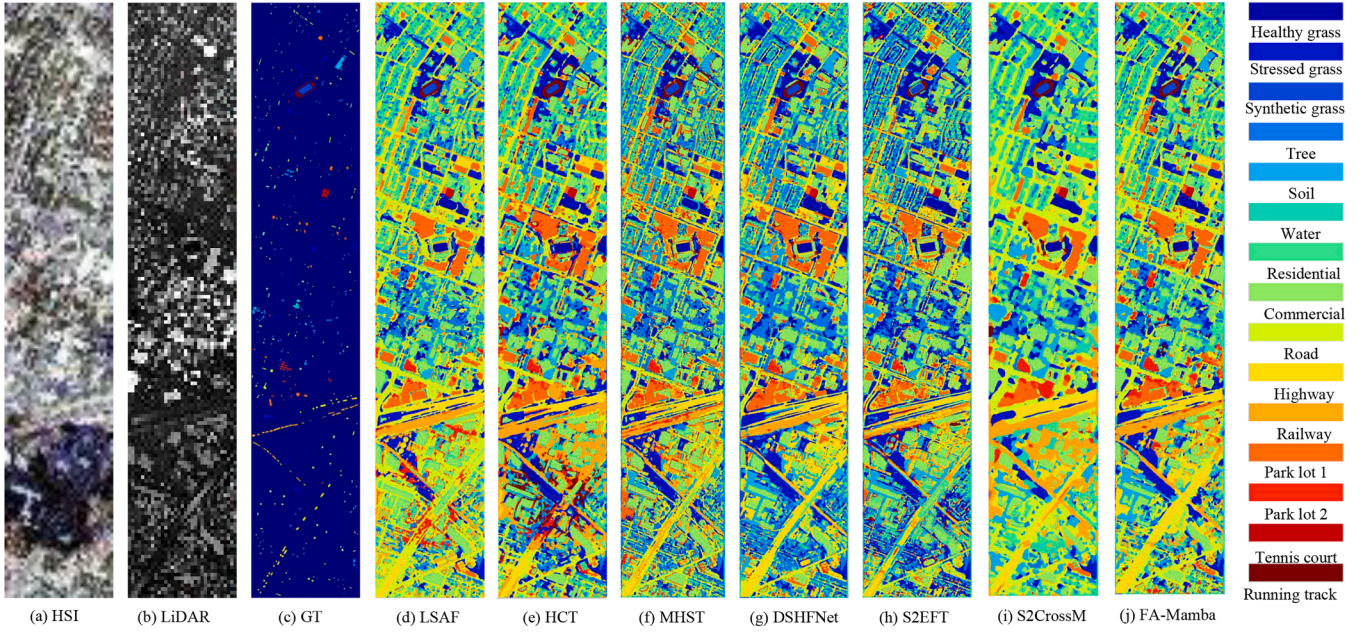


Fig. 3. Comparison of visual classification results of different methods on the Houston2013 dataset. (a) HSI, (b) LiDAR, (c) GT, (d) LSAF, (e) HCT, (f) MHST, (g) DSHFNet, (h) S2EFT, (i) S2CrossM, (j) FA-Mamba.

As shown in Fig. 1(b), one part of the features undergoes transformations through modules such as SS2D and adaptive frequency domain processing, acquiring characteristics modeled over long-range dependencies and highlighting key global frequency domain features. The other part of the features only passes through a simple SiLU function to introduce nonlinearity. Subsequently, these two sets of features are then combined through multiplication, completing the entire residual structure and generating the corresponding HSI and LiDAR features, Y_H and Y_L .

3.3. Coordinate fusion attention

Coordinate Attention (CA) effectively enhances model focus on critical regions by embedding positional information through decomposed spatial pooling operations. While demonstrating strong performance in channel-spatial relationship modeling, CA still exhibits certain limitations. CFA mechanism addresses these limitations through architectural improvements. Unlike conventional coordinate attention, CFA simultaneously models both short-range and long-range dependencies, thereby enhancing feature representation capabilities without significantly increasing computational overhead.

Specifically, traditional CA embeds positional information through channel dimension reduction followed by expansion, which risks feature degradation during channel compression. Algorithm 1 shows the full flow details of the mechanism. CFA preserves channel integrity by dividing the original channel dimensions into multiple groups. These groups undergo parallel processing through convolutional branches with varying kernel sizes. The parallel branches mainly consist of two parts: 1) A convolution, which is used to extract local detail features and supplement high-frequency information. The generated feature is denoted as X_1 . 2) Two global average pooling (GAP) operations along the horizontal and vertical directions, which capture long-range dependencies and retain precise positional information along these axes. The pooled features, X_w and X_h , are merged into X_2 , which then interacts with X_1 across dimensions. The interdimensional interaction fully leverages the characteristics of both branches. The local detail feature X_1 and the global pooled feature X_2 first enhance their own representations by incorporating complementary information from each other before un-

Algorithm 1 Coordinate Fusion Attention (CFA).

Require: Input feature map $X \in \mathbb{R}^{C \times H \times W}$

1: Group number n

Ensure: Output feature map $Y \in \mathbb{R}^{C \times H \times W}$

2: // Step 1: Group-wise processing

3: $X_n \leftarrow \text{Reshape}\left(X, \left[b \times n, \frac{C}{n}, H, W\right]\right)$ ▷ Eq. (19)

4: // Step 2: Local detail branch

5: $X_1 \leftarrow \text{Conv}(X_n)$

6: // Step 3: Global context branch

7: $X_w \leftarrow \frac{1}{H} \sum_{i=1}^H X_n(:, i, :)$ ▷ Horizontal GAP

8: $X_h \leftarrow \frac{1}{W} \sum_{j=1}^W X_n(:, :, j)$ ▷ Vertical GAP

9: $X_2 \leftarrow \text{Concat}(X_w, X_h)$ ▷ Global feature fusion

10: // Step 4: Interdimensional interaction

11: $A \leftarrow \sigma(X_1) \boxtimes X_2$ ▷ Enhance global with local

12: $B \leftarrow \sigma(X_2) \boxtimes X_1$ ▷ Enhance local with global

13: // Step 5: Dual-branch fusion

14: $F \leftarrow \Phi(A, B)$ ▷ Feature fusion function Φ

15: // Step 6: Feature recalibration

16: $Y_n \leftarrow X_n \odot F$ ▷ Element-wise multiplication (Eq. 18)

17: // Step 7: Output reconstruction

18: $Y \leftarrow \text{Reshape}(Y_n, [b, C, H, W])$ **return** Y

dergoing feature fusion. This results in a global spatial representation that preserves positional information. The process can be expressed as:

$$Y = X_n \odot \left[\underbrace{\Phi\left(\sigma(X_1) \boxtimes X_2, \sigma(X_2) \boxtimes X_1\right)}_{\text{Dual-branch Feature Fusion}} \right], \quad (18)$$

and

$$X_n = \text{Reshape}\left(X, \left[b \times n, \frac{C}{n}, h, w\right]\right), \quad (19)$$

where Φ is the dual-branch feature fusion function used to integrate the features from both branches; $\boxtimes$ represents interdimensional interaction, which enhances the details of one dimension using features from the other dimension; and σ is the normalization operation. The CFA

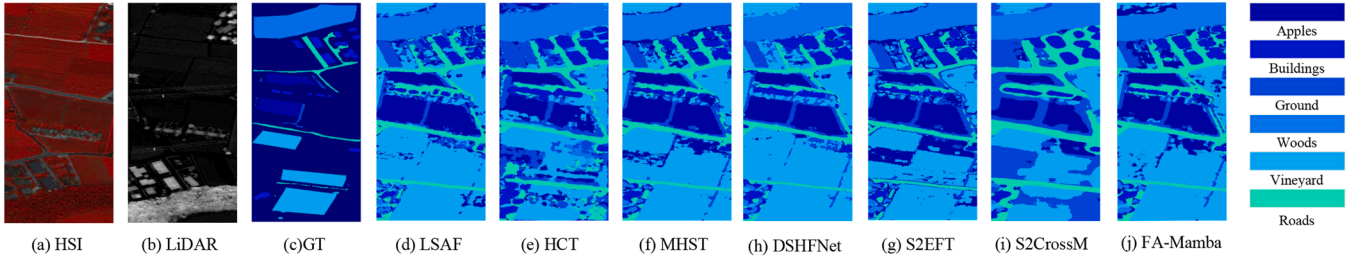


Fig. 4. Comparison of visual classification results of different methods on the Trento dataset. (a) HSI, (b) LiDAR, (c) GT, (d) LSAF, (e) HCT, (f) MHST, (g) DSHFNet, (h) S2EFT, (i) S2CrossM, (j) FA-Mamba.

Table 1

Detailed information of the three datasets.

Dataset	Houston2013		Trento		MUUFL	
	HSI	LiDAR	HSI	LiDAR	HSI	LiDAR
Image Size	349 × 1905	349 × 1905	600 × 166	600 × 166	325 × 220	325 × 220
Number of Bands	144	1	63	1	64	1
Wavelength Range	0.38-1.05μm	/	0.42-0.99μm	/	375-1050nm	/
Spatial Resolution	2.5m	2.5m	1m	1m	0.54 × 1.00m	0.60 × 0.78m
Number of Categories	15	15	6	6	11	11

module is strategically positioned after the FAM module to leverage FAM module's local spatial feature extraction capabilities, while CFA complements this with global spatial modeling, enabling optimal performance synergy between modules.

4. Experiment

4.1. Data description

In the experiments, three widely adopted multimodal remote sensing datasets, namely Houston (Debes et al., 2014), Trento, and MUUFL (Gader et al., 2013), are employed for evaluating classification performance. The Houston2013 Dataset, provided by the IEEE GRSS Data Fusion Contest, captures urban areas and includes 15 distinct land-cover classes, combining hyperspectral imagery (144 bands) with LiDAR data (1 band) at a spatial resolution of 2.5 m. The Trento Dataset, acquired in a rural region of southern Italy, comprises HSI (63 bands) and LiDAR (1 band) data at 1-meter resolution, and is annotated with six different land-cover types. The MUUFL Dataset, collected over a university campus environment, contains denoised HSI (64 bands) alongside dual-band LiDAR data, covering 11 categories of land cover. The diversity across these datasets in terms of spatial resolutions (ranging from 1 to 2.5 m), spectral dimensionality (from 64 to 144 bands), and land-cover complexity ensures a thorough and comprehensive evaluation of classification approaches. For more detailed information, please refer to Table 1.

4.2. Experimental settings

Evaluation Metrics: In terms of the performance evaluation of the model, we employed three commonly used metrics: Overall Accuracy (OA), Average Accuracy (AA), and Kappa (κ) to quantify the results. Where κ is used to measure the consistency between the classification results and random classification. The larger the value, the higher the consistency. Their calculation formulas are as follows:

$$OA = \frac{N_c}{N_a}, \quad (20)$$

$$AA = \frac{1}{C} \sum_{i=1}^C \frac{N_c^i}{N_a^i}, \quad (21)$$

$$\kappa = \frac{OA - P_e}{1 - P_e}, \quad (22)$$

$$P_e = \frac{1}{N_a^2} \sum_{i=1}^C N_r^i \times N_p^i, \quad (23)$$

where N_c and N_a represent the number of correctly classified samples and the total number of samples, respectively, while N_c^i and N_a^i correspond to N_c and N_a for each class. In κ , P_e is defined as the probability of hypothetical chance agreement.

Parameter Setting: We designed the entire network model based on the PyTorch platform and conducted training and testing on an NVIDIA A100 GPU with 80 GB of memory to meet computational requirements. The specific experimental setup is as follows: we used the Adam optimizer for parameter optimization, with an initial learning rate of 0.001 and a fixed batch size of 64. The model was trained for 200 epochs, with this number determined through a comprehensive evaluation of model convergence and performance stability across different datasets to ensure reliable results. To ensure a fair comparison among different algorithms, all experiments were conducted with the same batch triggering mechanism and parameter initialization strategy. GPU acceleration was achieved using torch.cuda device selection, and cross-entropy loss was used to optimize the multi-class classification task. Finally, a weighted fusion of the three branch outputs was employed to achieve feature integration at the decision level.

4.3. Comparisons with previous methods

We selected a set of advanced methods that are closely related to our task for comparison with FA-Mamba, particularly S2CrossMamba (Zhang et al., 2024), which also leverages an SSM architecture to enhance model performance. LSAF (Feng et al., 2021) proposes a plug-and-play linear self-attention module, which enhances the complementary features of HSI and LiDAR through a weighted fusion mechanism of channel attention and spatial attention. It also integrates an adaptive decision fusion module to aggregate classification results. HCT (Zhao et al., 2022) introduces a novel approach that combines hierarchical CNN and Transformer networks to fuse multi-source heterogeneous information. MHST (Ni et al., 2024) employs a multiscale spectral-spatial feature extraction module to capture detailed spatial and spectral information at different scales. DSHFNet (Feng et al., 2023) designs a dynamic scale hierarchical fusion network, which optimizes feature dimensions through a threshold-based dynamic multiscale feature selection mechanism, achieving hierarchical feature complementarity. S2EFT (Feng et al., 2024) addresses the limitations of Transformers in local

Table 2OA, AA and κ values on the Houston2013 dataset by considering HSI and LiDAR data.

Class	LSAF	HCT	MHST	DSHFNet	S2EFT	S2CrossM	FA-Mamba
Healthy grass	83.10	97.34	92.18	87.56	83.38	79.96	85.00
Stressed grass	83.08	96.62	96.80	99.34	97.09	94.36	99.91
Synthetic grass	100.00	84.75	97.82	98.81	97.23	97.21	98.22
Tree	89.49	96.78	99.81	99.81	98.86	94.98	97.63
Soil	100.00	100.00	100.00	99.91	99.72	99.53	100.00
Water	100.00	96.50	96.50	93.71	91.61	95.80	100.00
Residential	92.72	82.09	95.52	83.96	91.32	89.83	90.58
Commercial	92.69	95.54	96.39	80.63	90.03	94.30	91.17
Road	97.07	90.84	87.72	64.68	80.36	95.28	86.97
Highway	68.44	58.88	82.24	96.72	55.31	89.86	97.20
Railway	89.94	97.53	98.01	78.75	84.82	97.44	97.63
Parking lot 1	96.25	90.11	85.49	86.36	77.71	90.20	92.12
Parking lot 2	89.12	97.19	92.98	88.77	61.75	89.82	91.93
Tennis court	100.00	100.00	94.33	98.79	94.74	100.00	100.00
Running track	100.00	100.00	97.04	100.00	92.18	100.00	100.00
OA(%)	90.51	91.15	93.81	89.01	86.33	92.92	94.66
AA(%)	92.13	92.28	94.19	90.52	86.41	93.70	95.43
$\kappa(\times 100)$	89.69	90.40	93.28	88.06	85.18	92.32	94.20

Table 3OA, AA and κ Values on the Trento dataset by considering HSI and LiDAR data.

Class	LSAF	HCT	MHST	DSHFNet	S2EFT	S2CrossM	FA-Mamba
Apple trees	99.41	98.67	100.00	100.00	98.46	85.63	99.39
Buildings	87.51	99.92	99.78	98.92	97.95	98.78	99.46
Ground	91.71	99.73	94.92	79.95	89.30	94.39	93.05
Woods	99.14	99.96	99.80	99.58	100.00	100.00	99.97
Vineyard	99.89	67.39	99.83	97.69	96.83	80.46	100.00
Roads	95.22	86.57	84.76	96.56	89.18	92.37	89.78
OA(%)	97.84	86.96	98.21	98.35	97.23	90.25	98.71
AA(%)	95.48	92.04	96.52	95.45	95.29	91.94	96.94
$\kappa(\times 100)$	97.10	83.23	97.60	97.79	96.30	87.37	98.27

Table 4OA, AA and κ Values on the MUUFL dataset by considering HSI and LiDAR data.

Class	LSAF	HCT	MHST	DSHFNet	S2EFT	S2CrossM	FA-Mamba
Trees	97.00	97.03	98.86	95.06	97.37	97.09	98.24
Mostly grass	83.36	90.29	79.83	92.38	77.76	77.24	90.43
Mixed ground surface	90.79	90.07	79.63	72.90	83.13	92.44	91.27
Dirt and sand	94.18	94.18	93.95	94.87	82.54	76.94	98.27
Road	95.88	93.86	94.84	90.41	95.17	96.77	94.93
Water	94.58	95.71	92.78	1.81	92.10	70.88	96.84
Building shadow	87.36	87.09	87.13	97.97	78.97	81.19	86.61
Building	98.14	96.61	97.12	92.70	96.78	97.89	98.70
Sidewalk	75.91	46.35	68.77	56.08	63.22	80.24	43.16
Yellow curb	13.22	18.97	28.16	0.00	26.44	12.22	16.67
Cloth panels	84.77	75.39	93.75	0.00	91.02	92.58	92.79
OA(%)	93.70	92.95	92.43	88.65	91.17	92.65	94.15
AA(%)	83.20	80.50	83.17	63.11	80.41	79.68	82.55
$\kappa(\times 100)$	91.68	90.69	89.92	85.15	88.32	90.30	92.24

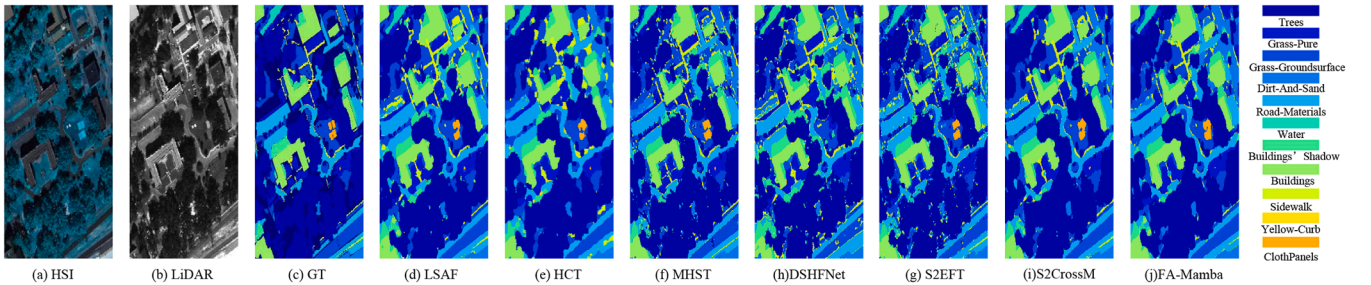
**Fig. 5.** Comparison of visual classification results of different methods on the MUUFL dataset. (a) HSI, (b) LiDAR, (c) GT, (d) LSAF, (e) HCT, (f) MHST, (g) DSHFNet, (h) S2EFT, (i) S2CrossM, (j) FA-Mamba.

Table 5
FLOPs and Params of different methods.

IMetric	ILSAF	IHCT	IMHST	IDSHFNet	IS2EFT	IS2CrossM	IFA-Mamba
FLOPs (M)	58.75	7.92	79.77	55.30	22.63	34.49	2.32
Params (M)	2.16	0.43	0.76	0.33	0.23	0.12	0.11

Table 6
Ablation study on frequency attention (F.A.), FAM and CFA.

Configuration	Houston2013			Trento			MUUFL		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Baseline	94.66	95.43	94.20	98.71	96.94	98.27	94.15	82.55	92.24
w/o F.A.	93.40	93.47	92.83	97.62	94.57	96.81	91.98	76.10	89.42
w/o FAM	92.26	92.68	91.60	97.04	93.87	96.05	91.70	76.64	88.96
w/o CFA	92.03	92.60	91.35	97.36	94.11	96.47	92.34	77.95	89.91

information processing by introducing a spatial information recognition module and a sliding group spectral embedding module. S2CrossMamba (Zhang et al., 2024) constructs an Inverted Bottleneck Cross-Modal module, which fuses spectral and elevation features using an addition operator and achieves cross-modal feature space recalibration using a multiplication operator. We evaluate the performance of these methods using a series of quantitative metrics, including per-class classification accuracy, OA, AA and κ . Additionally, we assess the computational efficiency of these methods using FLOPs and the parameters.

Comparison of Performance: The accuracy performance of FA-Mamba and these comparative methods on the Houston2013, Trento and MUUFL datasets can be found in Tables 2, 3 and 4. The best results are highlighted in bold. On the Houston2013 dataset, FA-Mamba demonstrates superior overall classification performance compared to other methods. It achieves the highest scores in OA, AA, and κ at 94.66%, 95.43%, and 94.20, respectively. Specifically, FA-Mamba improves OA by 0.85% over the suboptimal method MHST (93.81%), AA by 1.24% (MHST: 94.19%), and κ by 0.92 (MHST: 93.28). Among the 15 land-cover categories, FA-Mamba achieves the highest classification accuracy in 6 classes. Notably, FA-Mamba outperforms the sub-optimal methods in Highway and Stressed grass by 0.48% (DSHFNet: 96.72%) and 0.57% (DSHFNet: 99.34%), respectively. Furthermore, FA-Mamba achieves perfect 100% accuracy in Soil, Water, Tennis court, and Running track, highlighting its effectiveness in capturing high-frequency details and modeling long-range dependencies. On the Trento dataset, FA-Mamba achieves state-of-the-art performance with 98.71% OA and 98.27 κ , surpassing the suboptimal method DSHFNet (OA: 98.35%) by 0.36%. It demonstrates exceptional accuracy in critical categories such as Vineyard (100.00%), Apple Trees (99.39%), and Woods (99.97%), validating its robustness in agricultural scenarios. On the MUUFL dataset, FA-Mamba sets a new record with 94.15% OA and 92.24 κ , outperforming LSAF (OA: 93.70%) by 0.45%. The model excels in Buildings (98.70%), Dirt and Sand (98.27%), and Water (96.84%), but requires improvement in minority classes like Sidewalk (43.16%) and Yellow Curb (16.67%). The results highlight FA-Mamba's balanced capability in global spectral-spatial modeling and high-frequency detail extraction across both datasets. In addition, we generate visualized classification results, as shown in Figs. 3, 4 and 5, to further evaluate the classification effectiveness of each method.

Comparison of Computational Efficiency: Beyond classification accuracy, practical remote-sensing deployment also requires favorable computational efficiency. Therefore, we next compare FLOPs and parameter counts across methods. Table 5 presents a comprehensive computational efficiency comparison between FA-Mamba and comparative methods, where computational complexity is quantified through Floating Point Operations (FLOPs). As shown in Table 5, FA-Mamba (labeled as SST-Net in the table) demonstrates remarkable advantages in computational efficiency and model lightweighting. With only 2.32M FLOPs, it significantly reduces computational complexity by over 90% com-

pared to other methods (e.g., LSAF: 58.75M, MHST: 79.77M). In terms of parameters, FA-Mamba requires merely 0.11M, further compressing the model size by 8.3% compared to the suboptimal method S2CrossM (0.12M). These results indicate that FA-Mamba, through the synergistic and efficient design of state space models and frequency attention mechanisms, achieves consistently high classification accuracy (e.g., 94.66% OA on Houston2013) while drastically reducing computational resource demands, thereby making it well-suited for real-time processing of high-resolution remote sensing imagery.

4.4. Ablation study

To evaluate the effectiveness of key components in the proposed FA-Mamba, we conducted an ablation study on three datasets: Houston2013, Trento, and MUUFL. Specifically, we focus on two major aspects: the impact of FAM module and frequency attention therein, and the impact of the CFA mechanism. The experimental settings, results, and analyses are described in detail below.

Effects of FAM and frequency attention: The FAM module integrates SSM and frequency attention, capturing long-range dependencies of multimodal data in an efficient manner. The frequency attention adjusts features in the frequency domain via FFT, emphasizing key frequency characteristics. To assess their contribution to model performance, we compare the following three configurations: (1) Baseline: the full model, including FAM module with frequency attention; (2) Without frequency attention: retains FAM but removes the frequency attention; (3) Without FAM: removes the entire FAM module. We trained these configurations on Houston2013, Trento, and MUUFL datasets and recorded classification metrics: OA, AA, and Kappa coefficient (κ).

As shown in Table 6, removing the frequency attention led to a decrease in OA to 93.40%, 97.62%, and 91.98% on Houston2013, Trento, and MUUFL, respectively, demonstrating the importance of frequency feature enhancement by frequency attention. Removing FAM resulted in even larger performance drops of 2.40%, 1.67%, and 2.45%, highlighting the importance of FAM in long-range dependency modeling. The trends in AA and Kappa were consistent with those in OA, further validating the roles of FAM and frequency attention. Our analysis shows that frequency attention improves the model's ability to discriminate multimodal remote sensing features by enhancing task-relevant components in the frequency domain, while FAM provides a strong foundation for global context modeling through state space formulation. Together, these components significantly boost FA-Mamba's performance, especially on the class-imbalanced MUUFL dataset.

Effect of the CFA Mechanism: The CFA mechanism is designed to preserve global spatial characteristics and guide multimodal fusion through multi-scale feature extraction and adaptive cross-modal interaction prior to feature fusion. To evaluate its impact, we compared the following two configurations across Houston2013, Trento, and MUUFL:

(1) Baseline: the full model, including CFA; (2) Without CFA: with the CFA mechanism removed.

As observed in Table 6, removing CFA resulted in OA drops of 2.63%, 1.35%, and 1.81% on the three datasets, respectively, with similar trends in AA and Kappa. These results confirm that CFA plays a crucial role in enhancing model performance. By integrating multi-scale contextual information and preserving global spatial features, CFA improves the utilization of complementary HSI and LiDAR information, thereby increasing classification accuracy.

5. Conclusions

This paper proposes FA-Mamba, a novel multimodal remote sensing classification network that addresses the challenges of long-range dependency modeling and frequency-domain representation in high-resolution remote sensing imagery through the integration of SSM and frequency domain enhancement. FA-Mamba leverages the global modeling capability of SSM to efficiently capture long-range dependencies in multimodal data while maintaining linear computational complexity. By introducing the adaptive frequency domain enhancement, FA-Mamba further enhances the model's ability to adaptively adjust features in the frequency domain, highlighting critical frequency components essential for classification tasks. Additionally, to ensure comprehensive feature extraction, a CFA mechanism is incorporated to preserve global spatial characteristics and guide efficient multimodal feature fusion. In future research, we first plan to combine FA-Mamba with other advanced architectures, potentially further enhancing its ability to model complex spatial relationships in remote sensing data. Secondly, we will expand frequency-domain analysis to incorporate more sophisticated spectral filtering techniques. Lastly, we will apply FA-Mamba to other multimodal tasks to extend its versatility in the remote sensing field. A potential limitation is that FA-Mamba may still be less effective on rare or small classes under severe class imbalance, which motivates future exploration of imbalance-aware optimization and finer-grained boundary modeling.

CRedit authorship contribution statement

Danian Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Daixun Li:** Writing – review & editing, Visualization, Methodology; **Jitao Ma:** Writing – review & editing, Conceptualization; **Yibing Lu:** Conceptualization; **Yunsong Li:** Supervision; **Leyuan Fang:** Supervision; **Weiyang Xie:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors declare no conflicts of interest.

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